Theoretical Ecotoxicology
Session 5: Mixture modelling

University Duisburg-Essen SS 2026

Ralf B. Schäfer

Ecotoxicology, Faculty of Biology, UDE & RC One Health Ruhr

Intended learning outcomes of today's lecture

- Understanding and applying the major concepts for mixture toxicity estimation

- Knowledge on the predictive power of mixture models

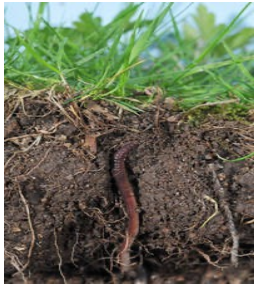
Learning targets and study questions

- Understanding and applying the major concepts for mixture toxicity
    - What is the *something from nothing* effect?
    - Explain the concepts of IA, CA and sum TU.
    - Discuss the choice between IA and CA.
    - What are antagonism and synergism?
- Knowledge on the predictive power of mixture models
    - Outline shortcomings of CA and IA.
    - How well do CA and IA match with observations, and how strong is the deviation?

Mixtures are abundant in ecosystems

76 of 82 target chemicals found in soils around Paris

Gaspéri et al. 2022 *Environ. Sci. Poll. Res.* 25: 23559



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398 of 2316 target chemicals found in Greek waste water samples

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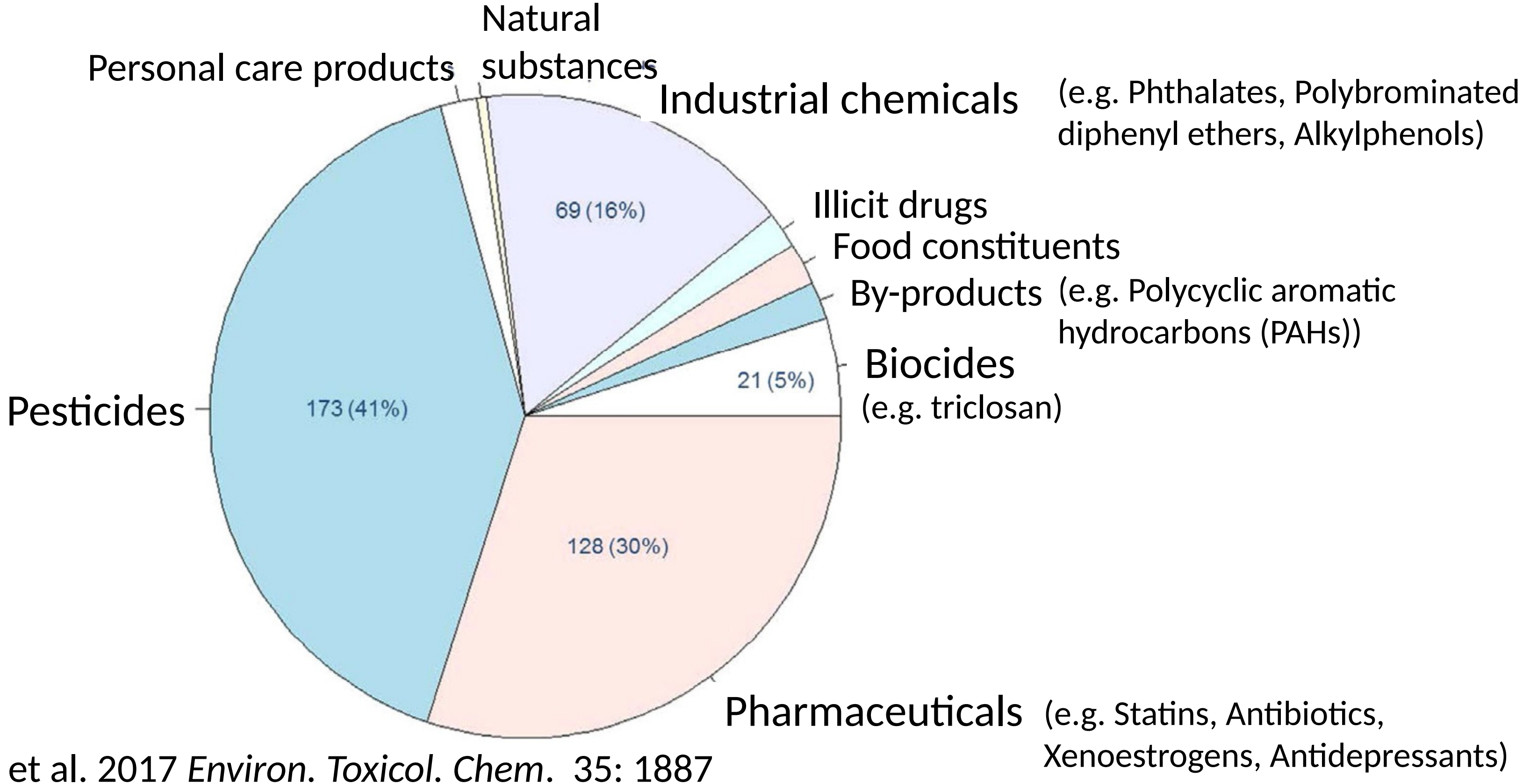
High diversity of modes of action

426 compounds found in 3 European river basins



Busch et al. 2017 *Environ. Toxicol. Chem*. 35: 1887

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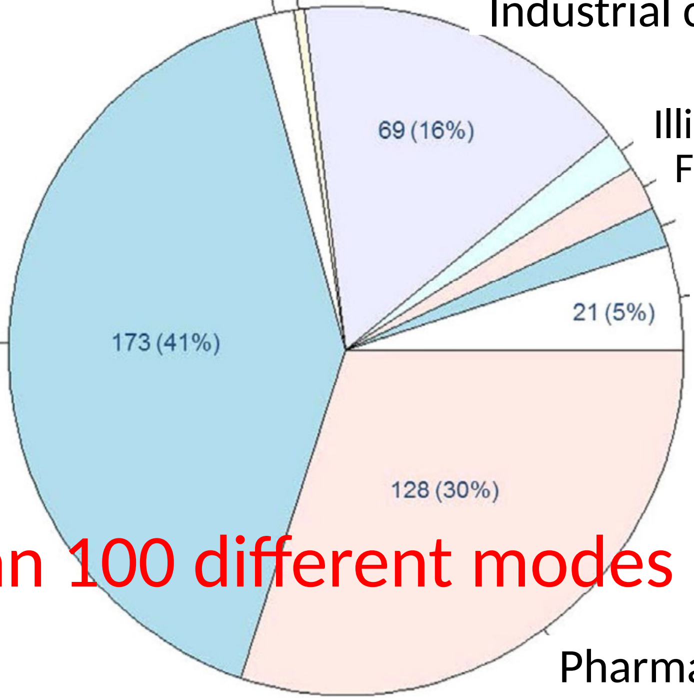
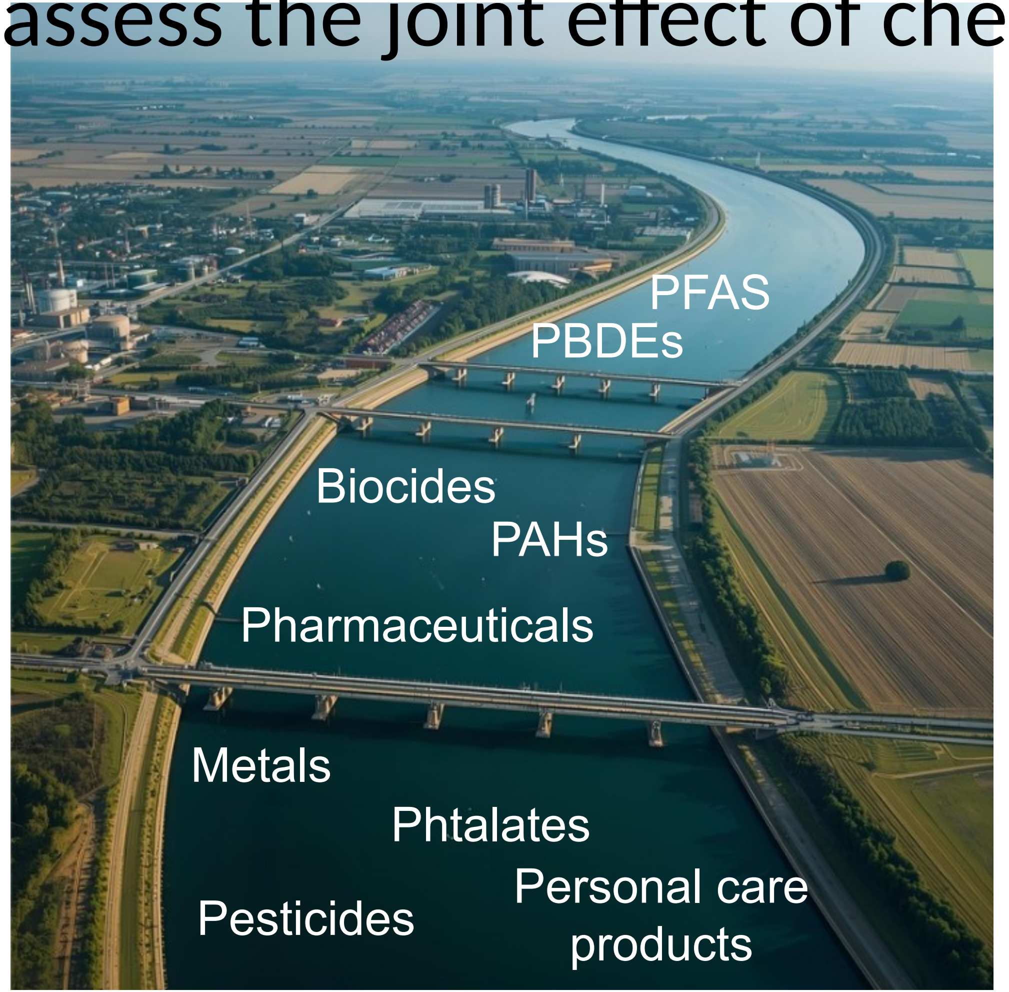


More than 100 different modes of toxic action

Busch et al. 2017 *Environ. Toxicol. Chem.* 35: 1887

How to assess the joint effect of chemicals?



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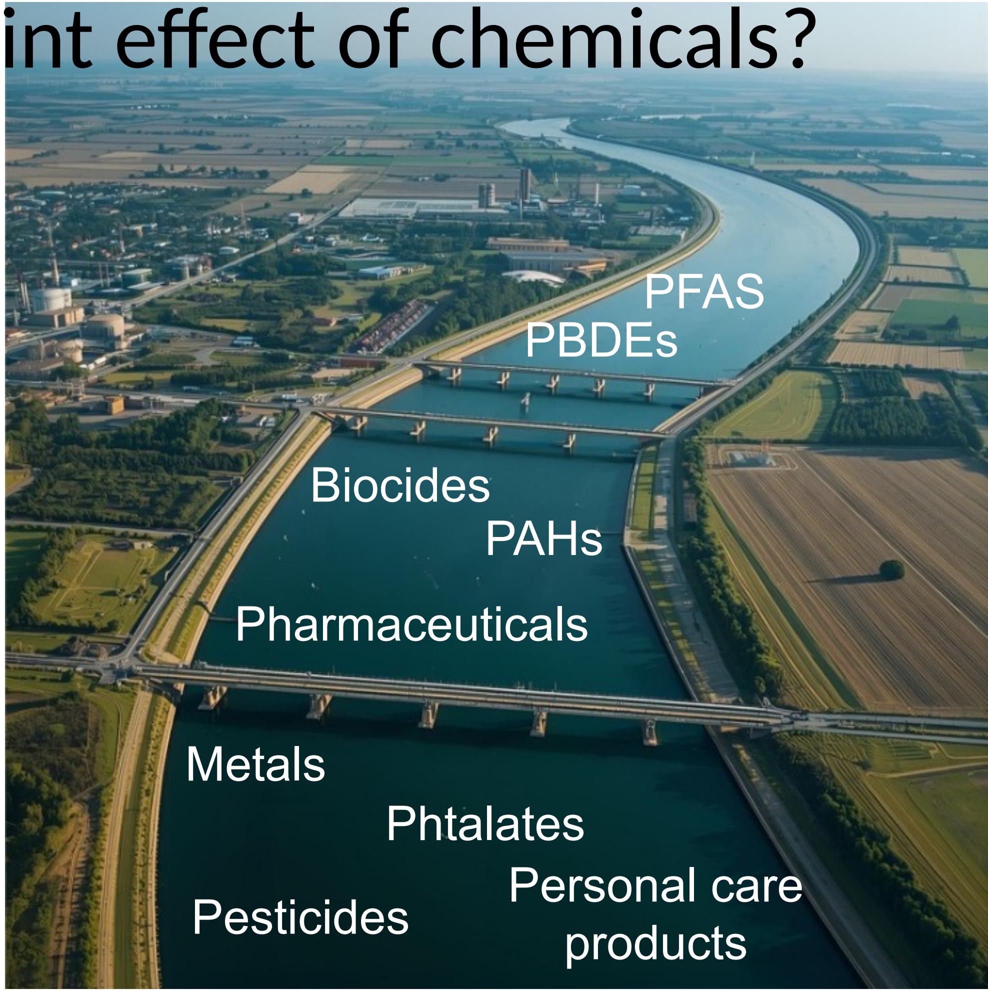
How to assess the joint effect of chemicals?

Imagine that we detected 20 chemicals at concentrations c_1, c_2 … c_{20}. What data would you need as a prerequisite to estimate their joint effect and how could you calculate a joint effect?

Discuss 5 minutes with your neighbour and then report to class.



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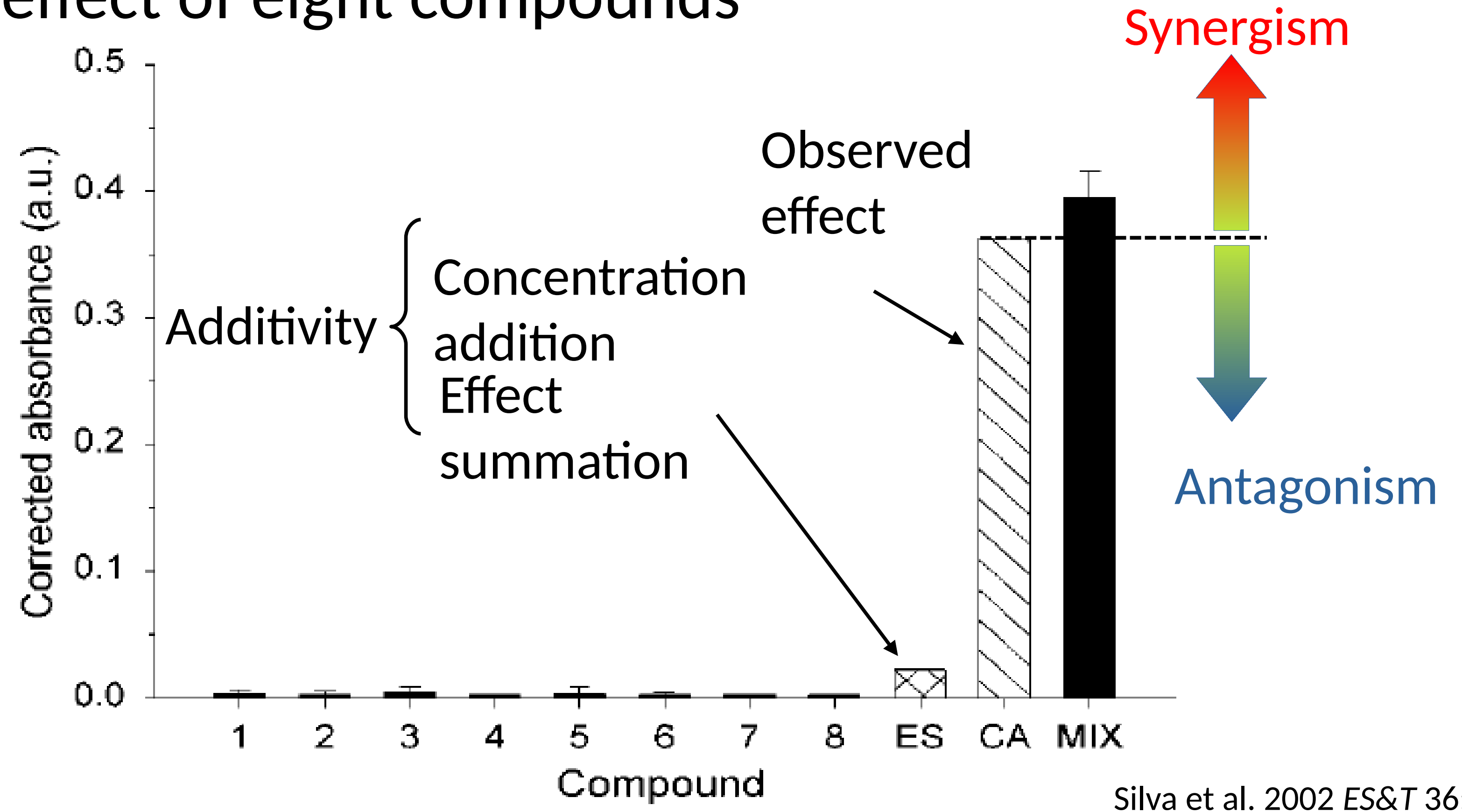
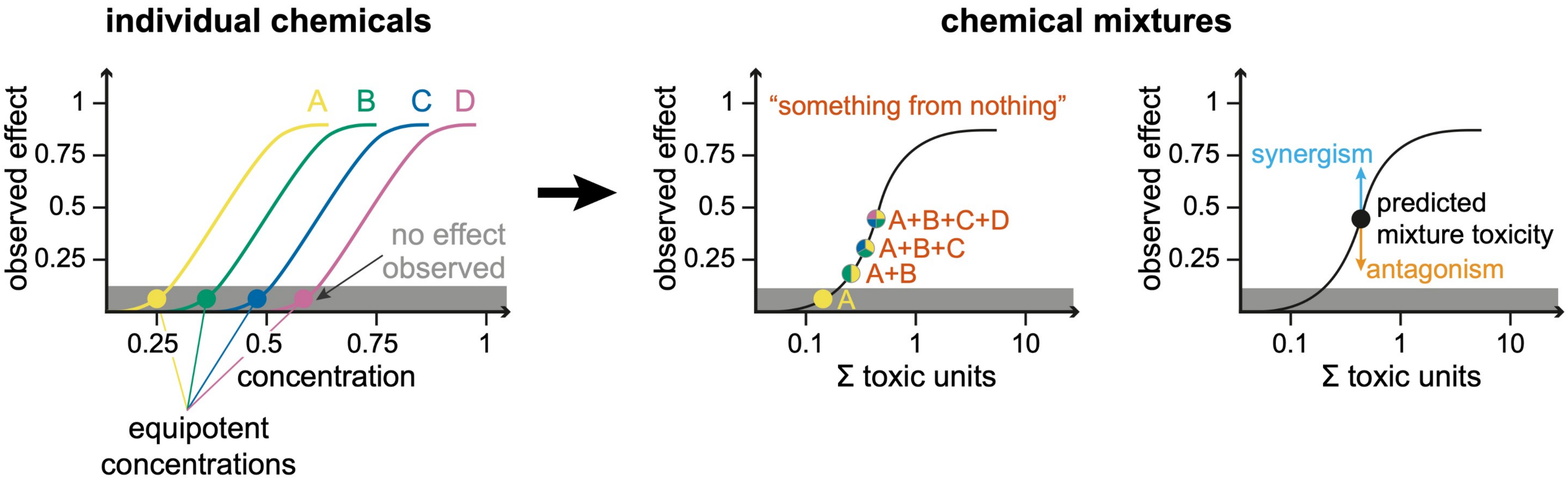
Mixture toxicity can matter!

Joint effect of eight compounds

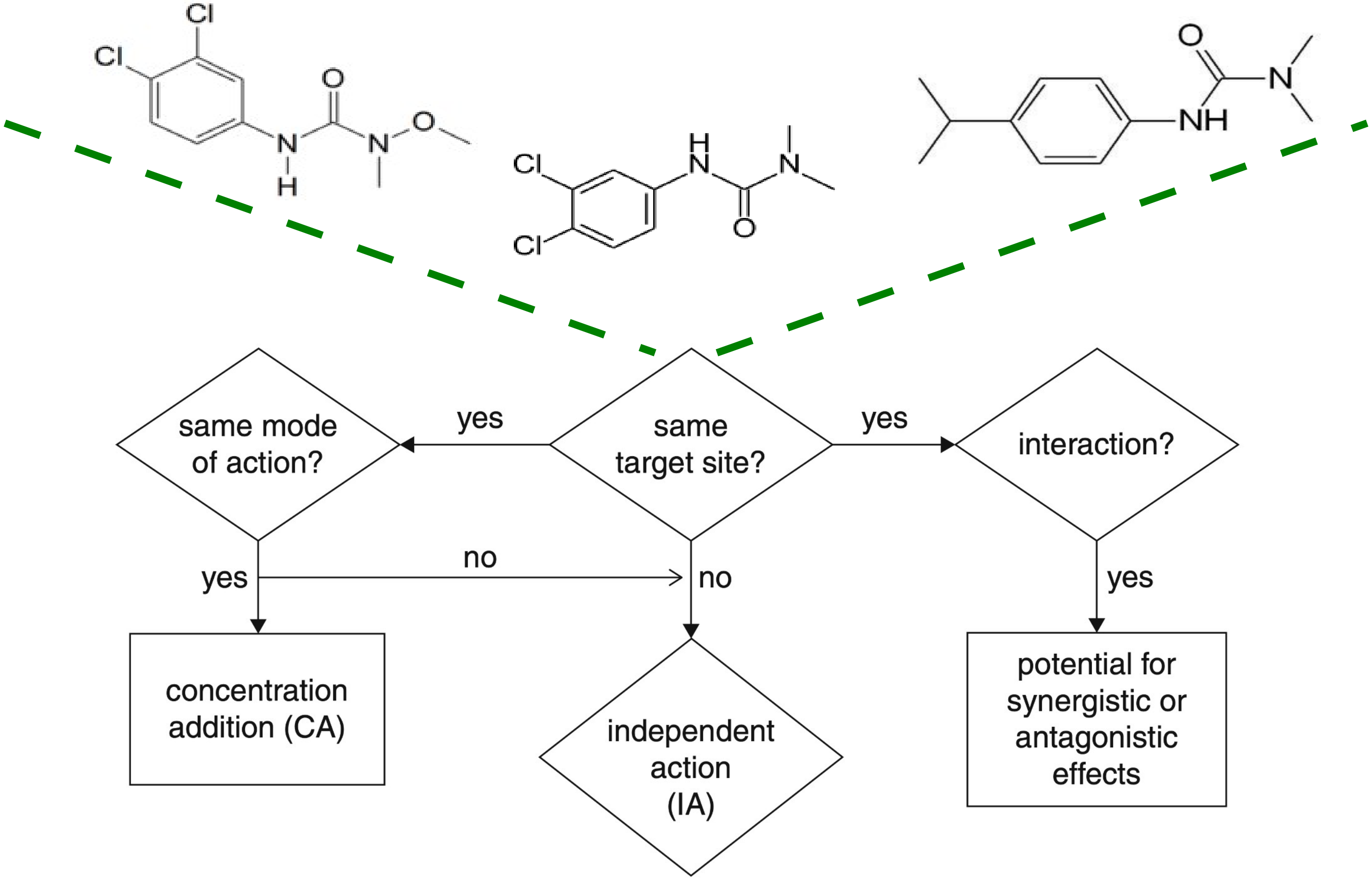


Silva et al. 2002 *ES&T* 36: 1751

Something from nothing effect



Mixtures and mode of action



Escher 2013

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^{n} \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

c_i: concentration of compound *i*

c_{mix}: concentration of the mixture

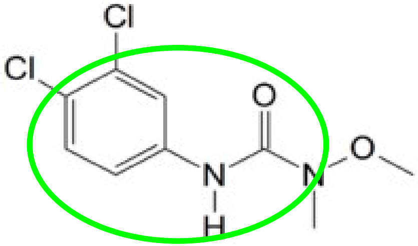
ECx_i: effect concentration x% for *i*

ECx_{mixCA}: mixture concentration having *x*% of effects following CA

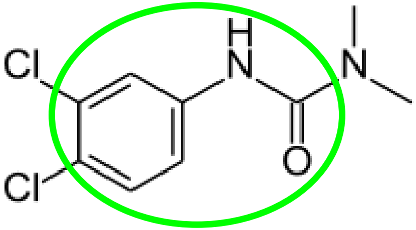
Chevre & Gregoria 2013

Example: Phenylurea herbicides

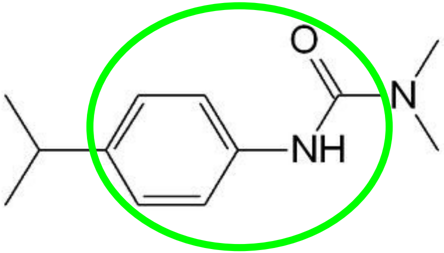
Diuron



Linuron



Isoproturon



Algae 72-h EC50, growth (µg L^{-1})	2.7	16	13

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Observed conc. (µg L^{-1})	5.4	160	1.3
p_i	$\frac{5.4}{166.7} = 0.032$	$\frac{160}{166.7} = 0.96$	$\frac{1.3}{166.7} = 0.008$
Algae 72-h EC50, growth (µg L^{-1})	2.7	16	13
$\frac{p_i}{EC_{xi}}$	$\frac{0.032}{2.7} = 0.012$	$\frac{0.96}{16} = 0.06$	$\frac{0.008}{13} = 0.0006$

Concentration addition (CA)

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$$\frac{p_i}{EC_{xi}} \qquad \frac{0.032}{2.7} = 0.012 \qquad \frac{0.96}{16} = 0.06 \qquad \frac{0.008}{13} = 0.0006$$

$$\sum_{i=1}^{n} \frac{p_i}{EC_{xi}}$$

0.012 + 0.06 + 0.0006 = 0.0726

$$ECx_{mixCA} = \frac{1}{0.0726} = 13.77$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^{n} \frac{c_i}{EC_{xi}}$$

c_i: concentration of compound *i*

EC_{xi}: effect concentration *x*% for compound *i*

Observed conc. (µg L^{-1})	5.4	160	1.3
Algae 72-h EC50, growth (µg L^{-1})	2.7	16	13

$$\frac{c_i}{EC_{xi}} \qquad \frac{5.4}{2.7} = 2 \qquad \frac{160}{16} = 10 \qquad \frac{1.3}{13} = 0.1$$

$$sumTU = 2 + 10 + 0.1 = 12.1$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^{n} \frac{c_i}{EC_{xi}}$$

c_i: concentration of compound *i*

EC_{xi}: effect concentration *x*% for compound *i*

$$sumTU = 2 + 10 + 0.1 = 12.1$$

Same as $$\frac{\sum_{i=1}^{n} c_i}{ECx_{mixCA}} = \frac{166.7}{13.77} = 12.1$$

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + ... + c_n) = 1 - \prod_{i=1}^{n} [1 - E(c_i)]$$

$E(c_{mixIA})$: predicted effect of the mixture following IA

$E_{(ci)}$: effect of compound *i* at concentration *c*

<u>Example: Insecticide, herbicide and fungicide</u>

	Diuron	Thiacloprid	Tebuconazole
Observed concentrations (µg L^{-1})	50	10	20

Van Wijngaarden 1996 *Environ. Toxicol. Chem.* 15:1133

Independent action (IA)

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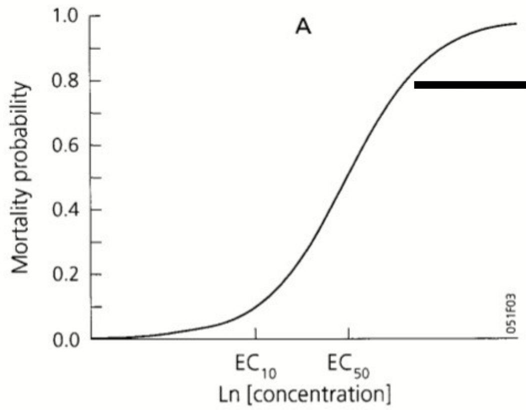


Derive $E(c_i)$

	Diuron	Thiacloprid	Tebuconazole
$E(c_i)$	0.05	0.2	0.35

Van Wijngaarden 1996 *Environ. Toxicol. Chem.* 15:1133

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \ldots + c_n) = 1 - \prod_{i=1}^{n} [1 - E(c_i)]$$

$E(c_i)$	0.05	0.2	0.35
$1- E(c_i)$	0.95	0.8	0.65

$$\prod_{i=1}^{n} [1 - E(c_i)]$$

0.95 * 0.8 * 0.65 = 0.49

$$E(c_{mixIA}) = 1 - 0.49 = 0.51$$

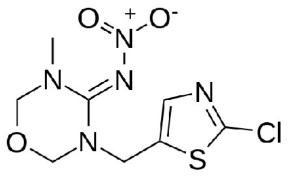
Computation of IA often hampered by data availability (mostly only NOEC, LOEC, EC50 reported)

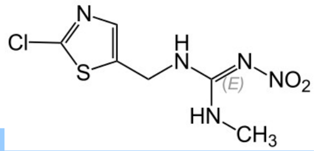
Chevre & Gregoria 2013

Exercise

$$E(c_{mixIA}) = E(c_1 + c_2 + \ldots + c_n) = 1 - \prod_{i=1}^{n} [1 - E(c_i)] \qquad sumTU = \sum_{i=1}^{n} \frac{c_i}{EC_{xi}}$$

Please calculate the mixture toxicity using CA as sumTU and IA models for the following situation:

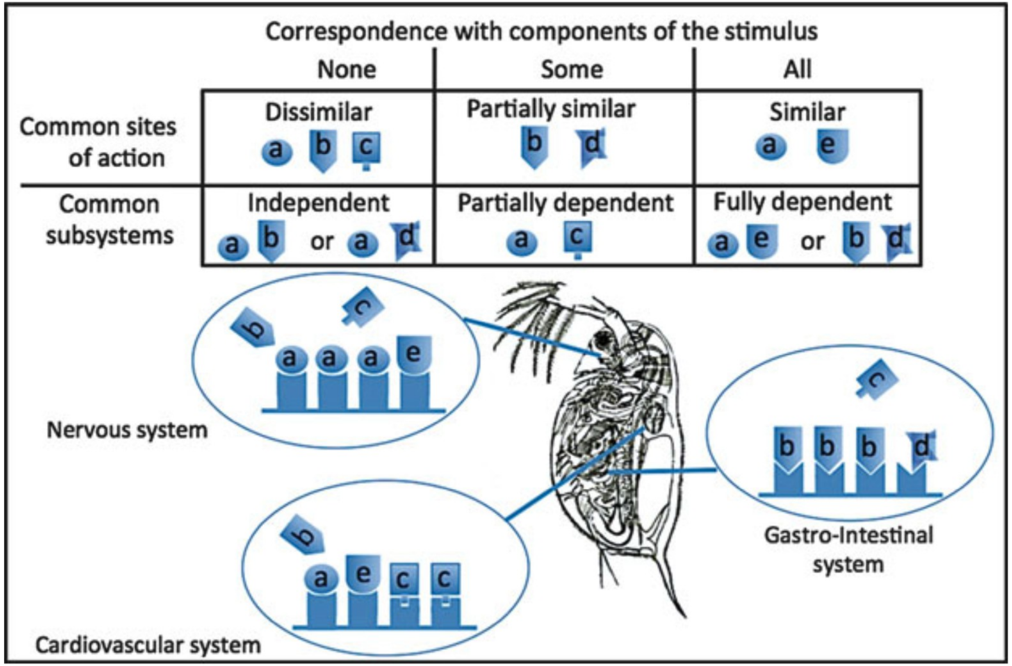




	Thiametoxam	**Clothianidin**
96-h LC50 Chironomus riparius (µg/L)	100	29
Observed concentrations (µg/L)	5	2.9
$E(c_i)$	0.02	0.2

CA or IA?

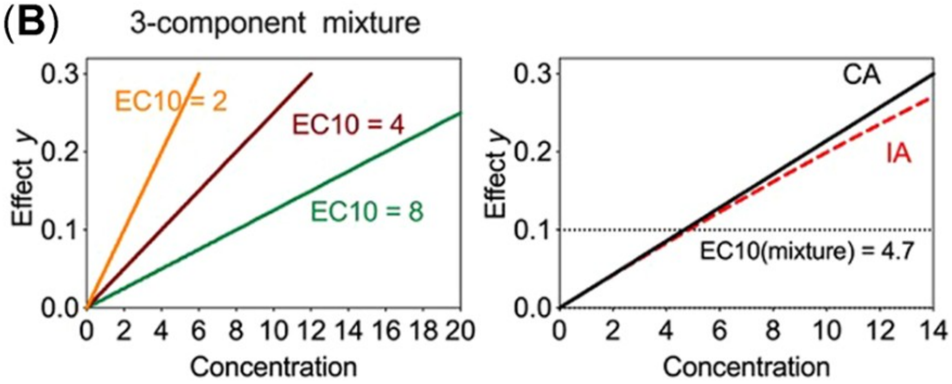
- Same target site and MoA → CA
- But MoA can vary between organs

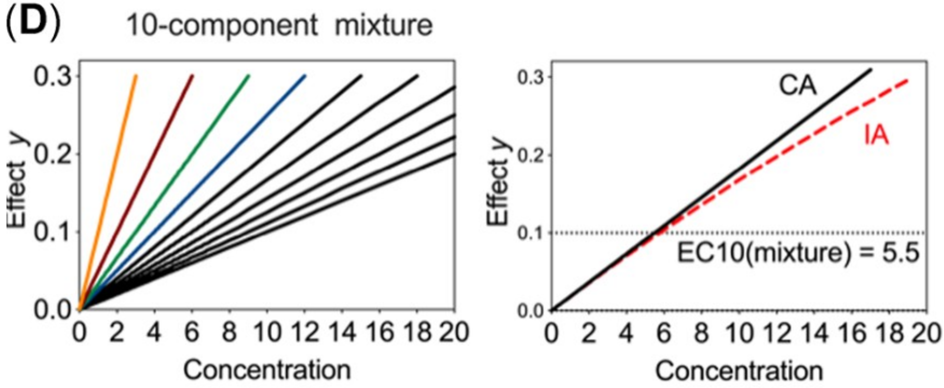


Chevre & Gregoria 2013

CA or IA?

- For equipotent mixtures with low effect levels (i.e. <EC10), often predictions converge
- CA standard because data often too limited for IA

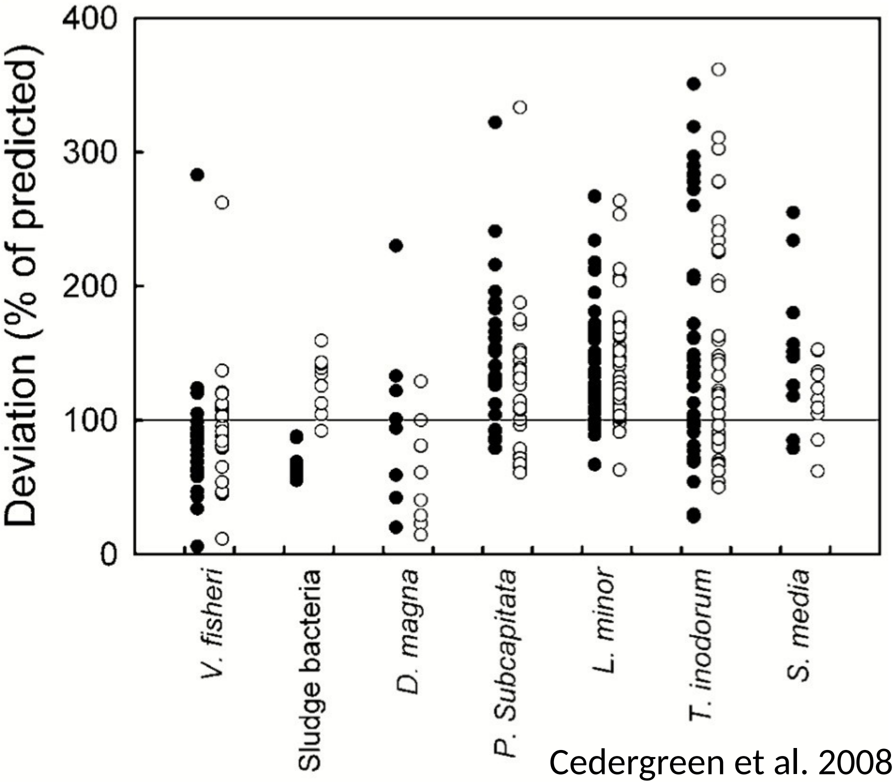




Escher et al. 2020 *Environ. Toxicol. Chem.* 39:2552

How good is the prediction?

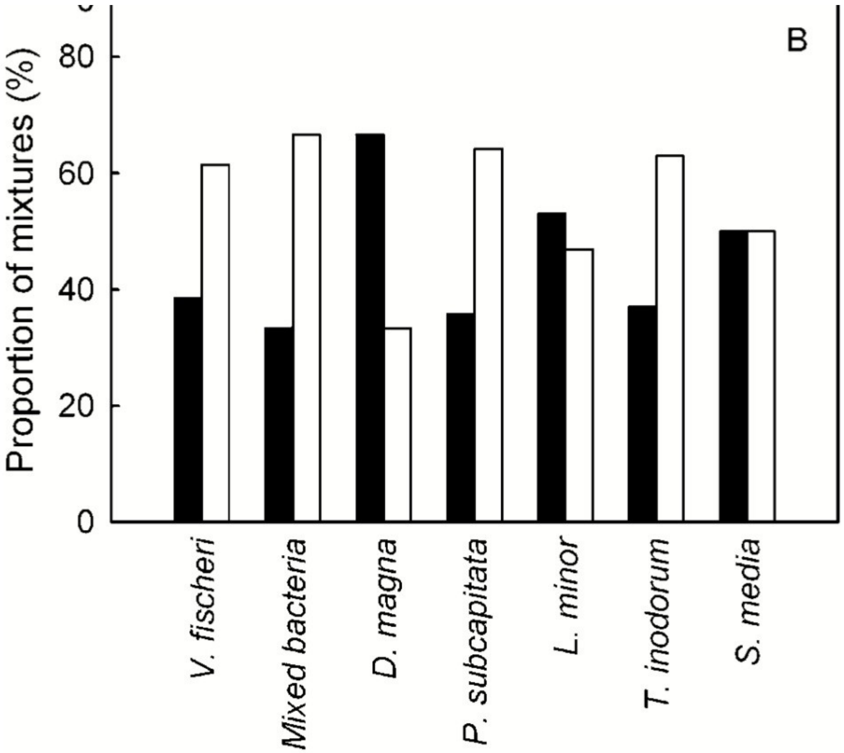
Ratio of CA (black) and IA (white) prediction to observed EC50



Cedergreen et al. 2008 *Environ. Toxicol. Chem.* 27(7): 1621

How good is the prediction?

Proportion of mixtures better described with CA (black) and better described with IA (white)



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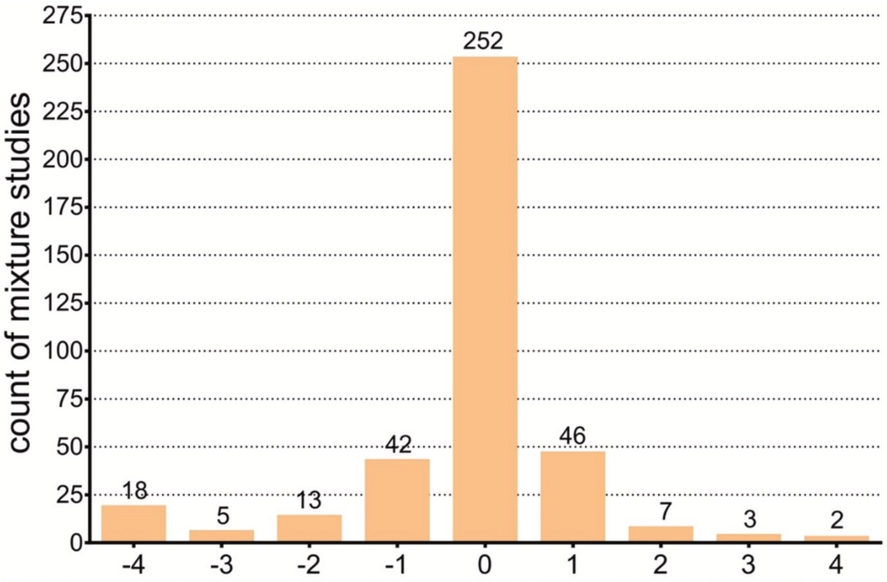
How good is the prediction?

Observations largely within 1 order of magnitude of mixture prediction



Class	-4	-3	-2	-1	0	1	2	3	4
IPQ range	<- 700%	-700% - -500%	-500% - -300%	- 300% - -100%	-100% - 100%	100% - 300%	300% - 500%	500% - 700%	>700%
	synergisms					antagonism			

Martin et al. 2021 *Environ. Intern.* 146: 106206

How to identify synergism and antagonism?

- Model Deviation Ratio (MDR) often used in Ecotoxicology:

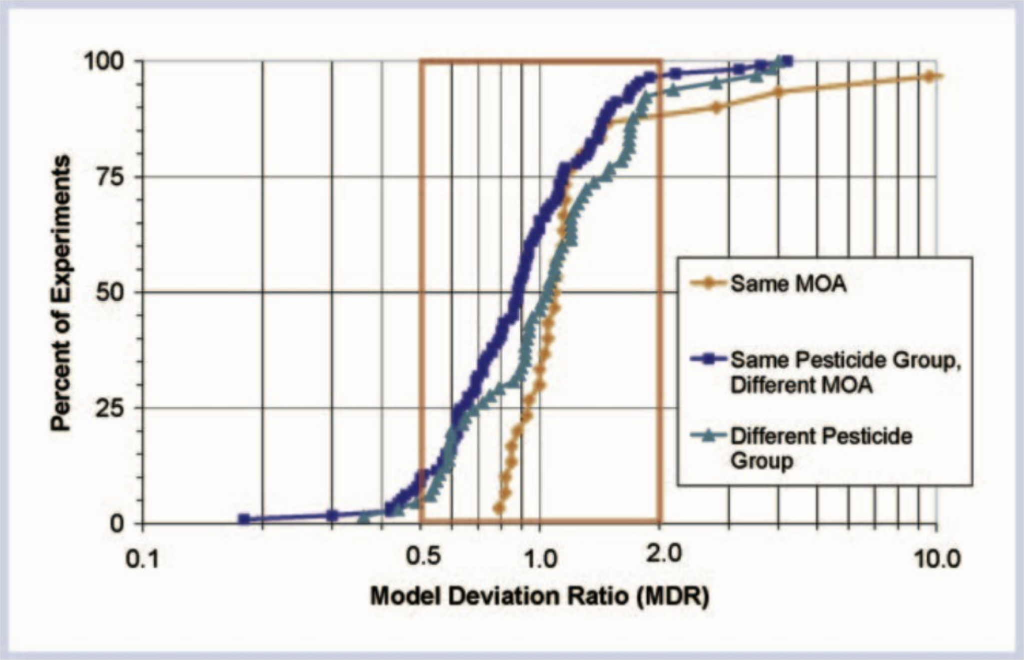
$$\text{MDR} = \frac{\text{Expected/predicted EC with null model}}{\text{Observed EC}}$$

- MDR sets threshold based on effect size (e.g. 0.5 and 2)
- Why not statistical significance?
  - Sample-size dependent: e.g. low effect size (deviation from model) significant for large sample size, high effect size not significant for small sample size

Belden et al. 2007 Integr. *Environ. Assess. Manag. 3* (3): 364

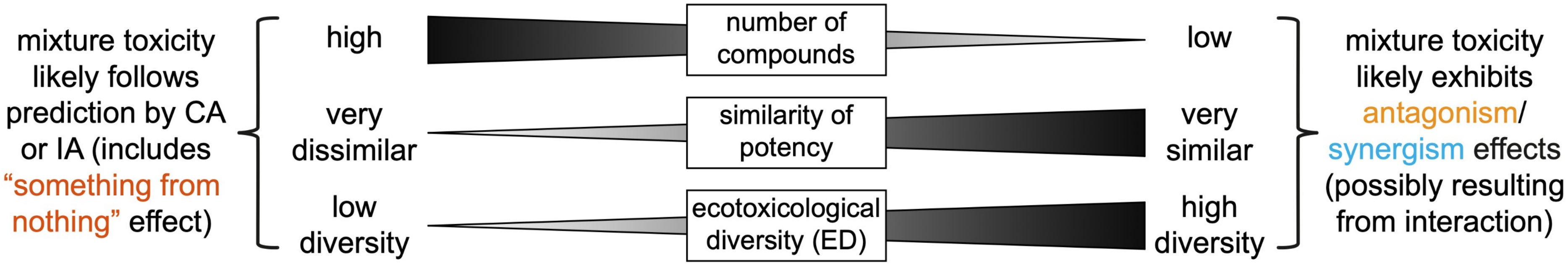
How good is the prediction?

MDR for CA for pesticide mixtures → Reasonable match for majority of experiments, low fraction of synergism/antagonism



Belden et al. 2007 Integr. *Environ. Assess. Manag. 3* (3): 364

What controls the frequency of synergism and antagonism?

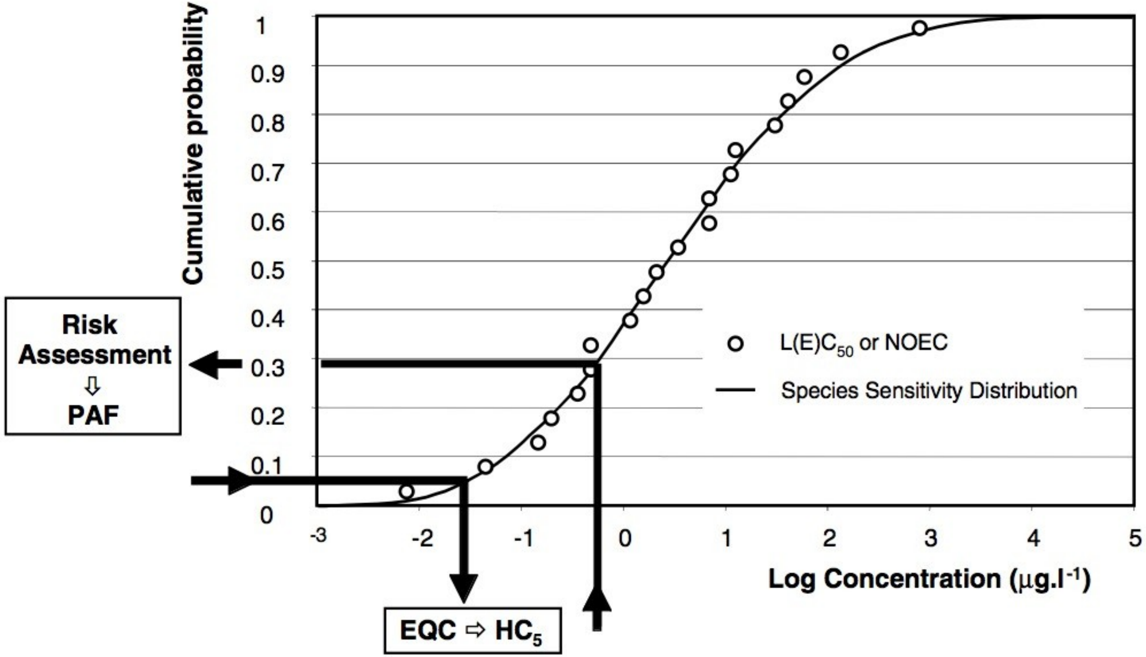


msPAF: Mixture prediction for SSDs

- Mixture prediction (CA and IA/RA) for community level effects
- Instead of *ECx* and *E(c)* values, use *Hcp* and *PAF* values from single-substance SSDs

SSDs for risk assessment

- Hazardous concentration (HC_p): conc. at which *p*% affected
- Potentially affected fraction PAF: given by concentration



Cutoff for community PNEC

Posthuma, Traas & Suter 2002

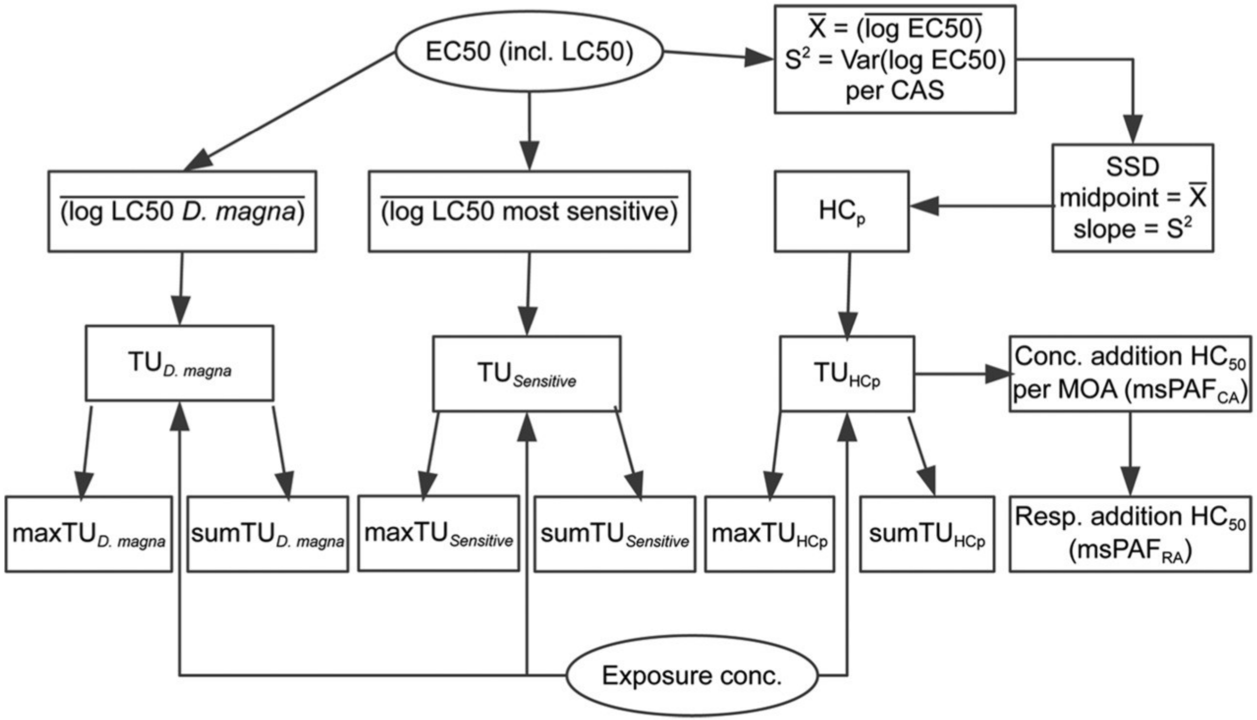
msPAF: Mixture prediction for SSDs

- Mixture prediction (CA and IA/RA) for community level effects
- Instead of ECx and $E(c)$ values, use Hcp and PAF values from single-substance SSDs
- $msPAF_{CA}$: Aggregate Hcp for all compounds with same MoA
- Alternative: $sumTU_{HCp}$

$$msPAF_{RA} = 1 - \prod_{i=1}^{n} \left(1 - msPAF_{CA,i}\right)$$

Example for community-level mixture indices

IA/RA was only calculated in SSD context due to lack of data



Schäfer et al. 2013 *Environ. Sci. Technol.* 47 (14): 7996

Example for community-level mixture indices

For 4 of 5 data sets $sumTU_{HC5}$ best metric, but maxTU and metrics based on *D. magna* or most sensitive test species not much worse

Exposure metric	Rank sum of metric across countries	Number of times metric was among models within a range of 2 to BIC of best-fit model
sumTU_{HC5}	14	4
$\mathrm{sumTU}_{Sensitive}$	18	3
maxTU_{HC5}	21	3
$\mathrm{maxTU}_{Sensitive}$	21	3
$\mathrm{maxTU}_{D.magna}$	22	2
$\mathrm{sumTU}_{D.magna}$	26	3
$\mathrm{msPAF}_{\mathrm{RA}}$	32	1
maxTU_{HC50}	35	1
sumTU_{HC50}	36	1

Schäfer et al. 2013 *Environ. Sci. Technol.* 47 (14): 7996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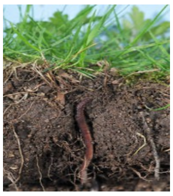

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High diversity of modes of action

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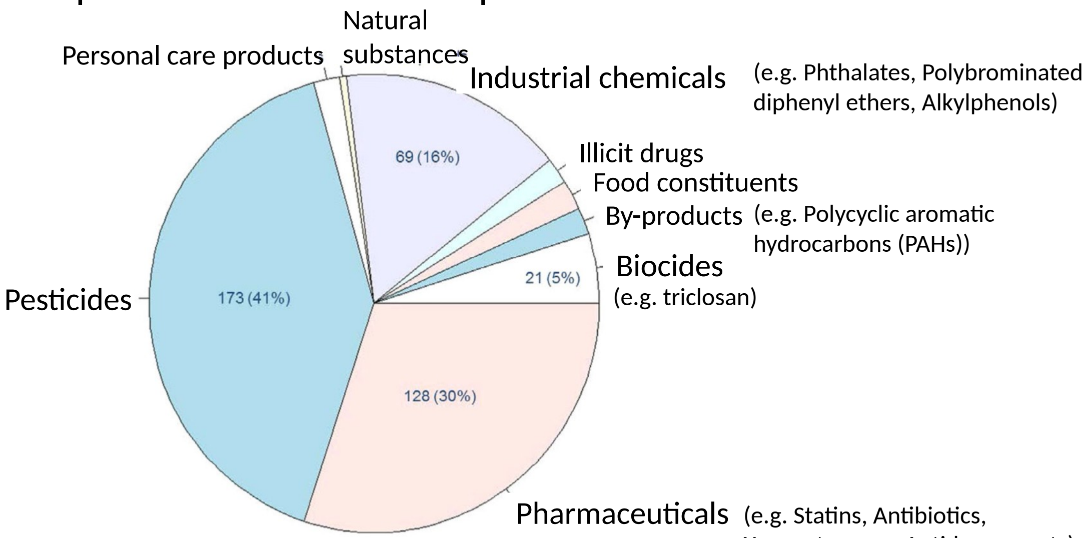

5 Busch et al. 2017 *Environ. Toxicol. Chem*. 35: 1887

The study found 426 compounds, although the real number would be higher given that legacy chemicals such as PCBs or metals were not measured.

High diversity of modes of action

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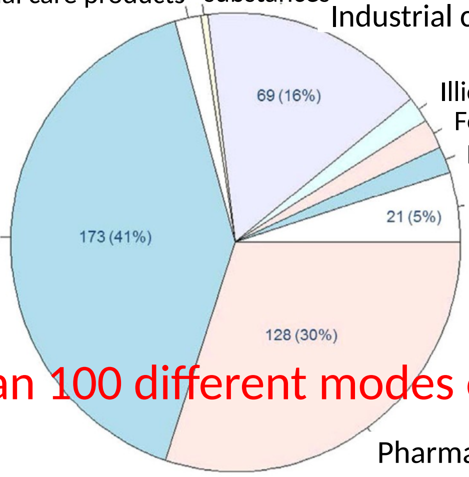

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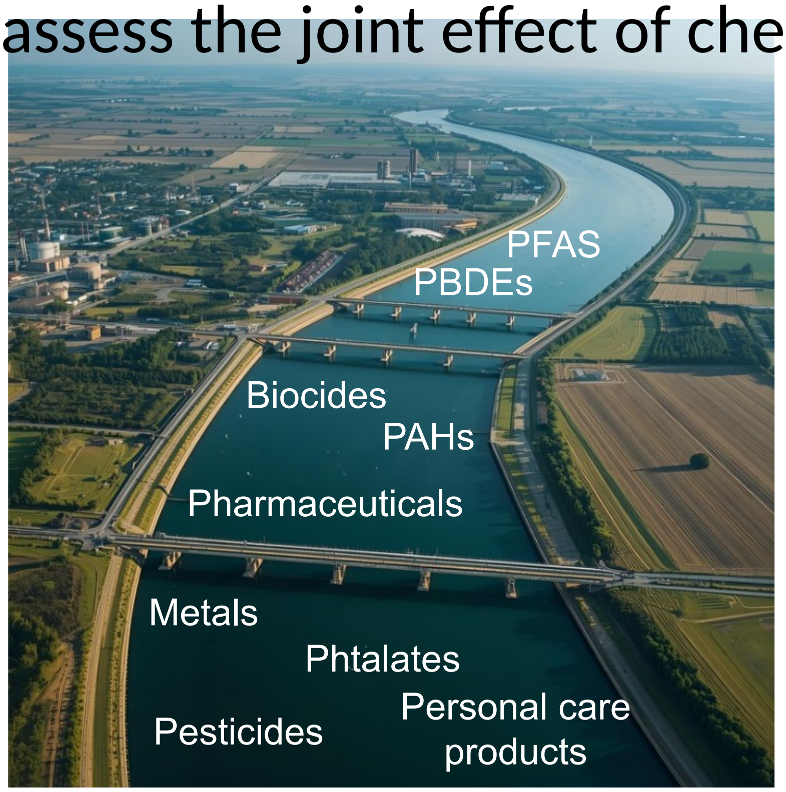
How to assess the joint effect of chemicals?



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How to assess the joint effect of chemicals?

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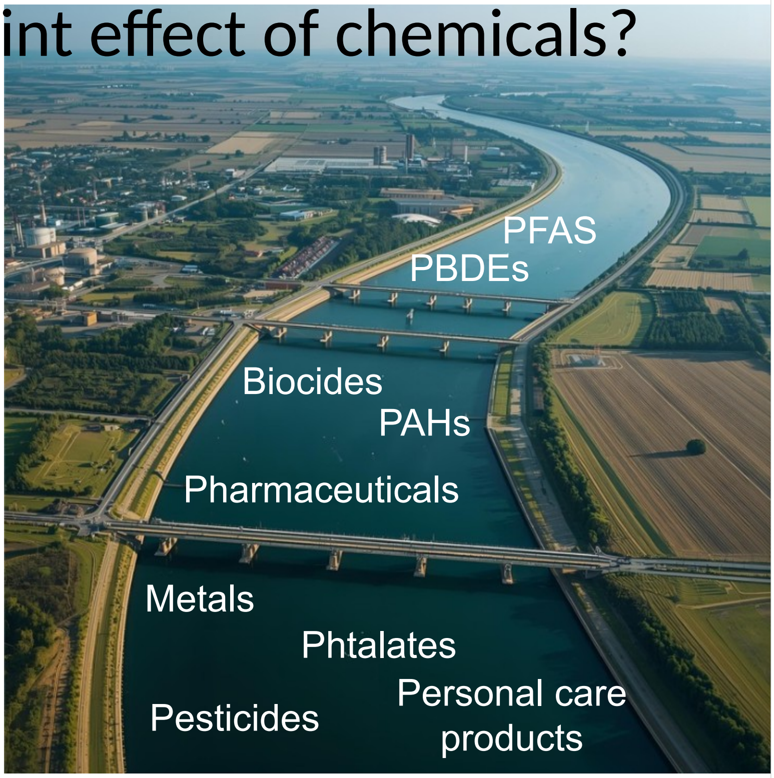
Discuss 5 minutes with your neighbour and then report to class.



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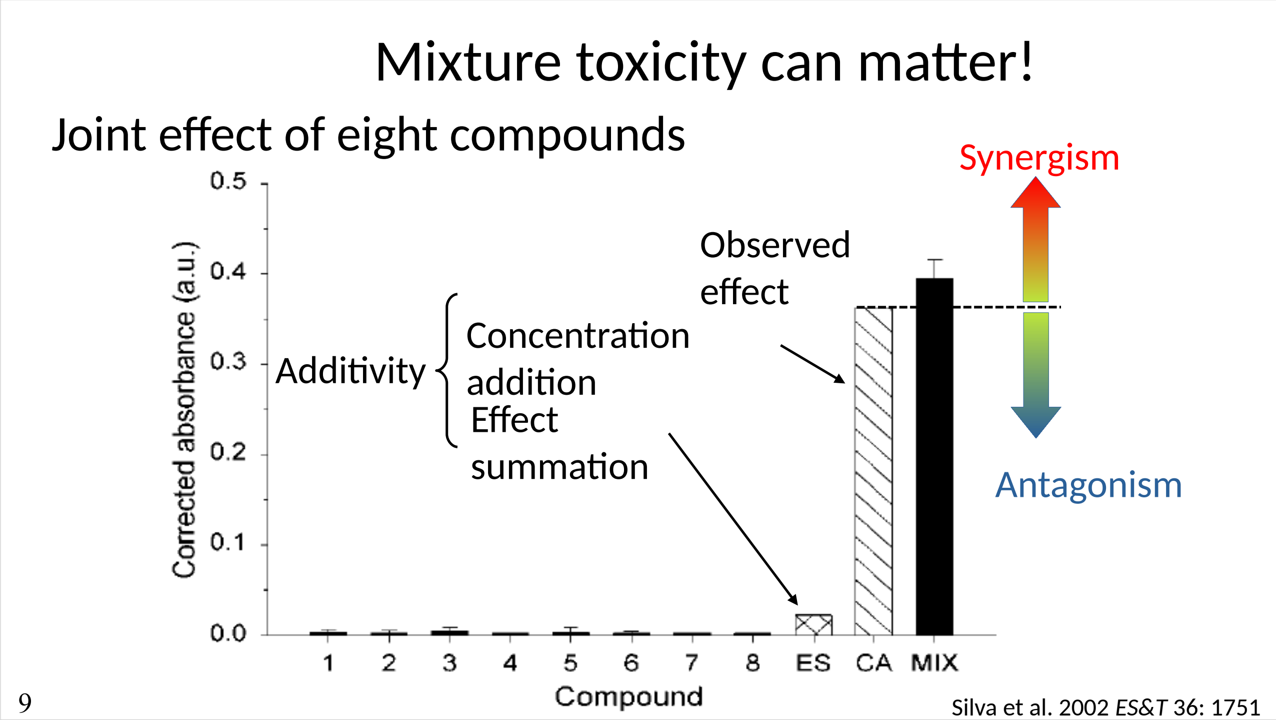
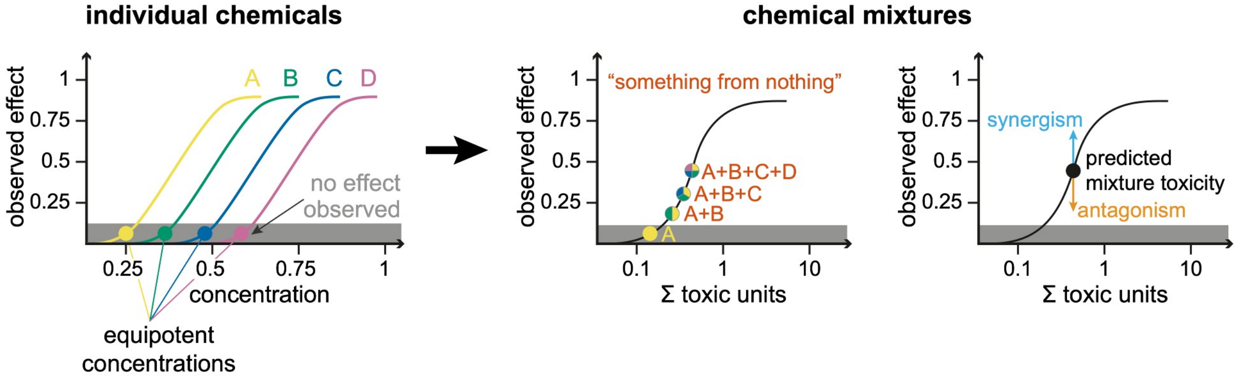


Effect summation is also termed Independent Action.

Antagonism: Mixture toxicity is smaller than expected (i.e. smaller than predicted by a null model, here the concentration addition model)
Synergism: Mixture toxicity is larger than expected (i.e. larger than predicted by a null model)
Additivity: Toxicity of mixture can be predicted by a null model, here CA or IA/ES.

Something from nothing effect



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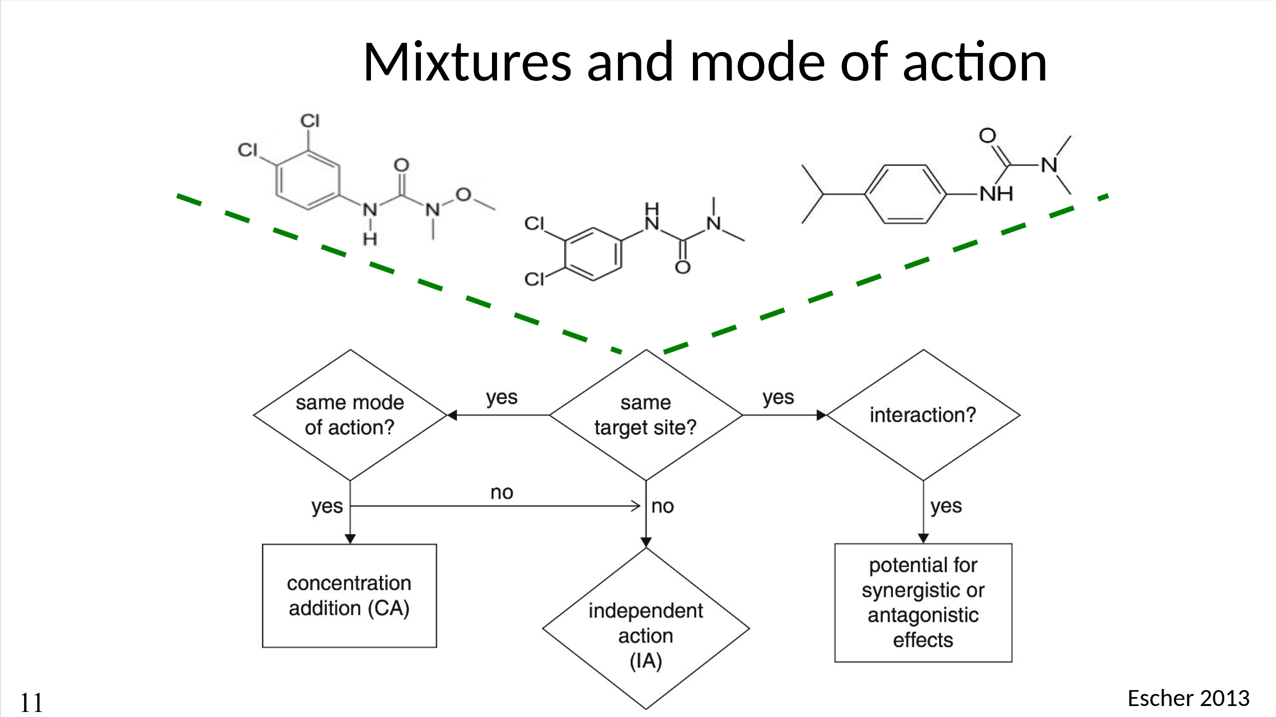
The *something from nothing* effect is only a consequence of adding up concentrations benchmarked by their toxicity, it does not imply mechanistic interactions.



Escher B. 2013: Modes of Action of Chemical Pollutants. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379983

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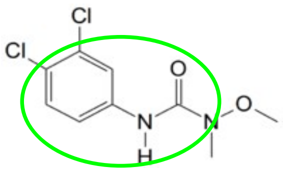
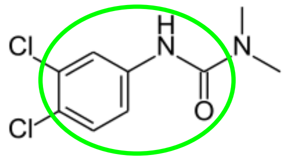
Example: Phenylurea herbicides

Chevre & Gregoria 2013

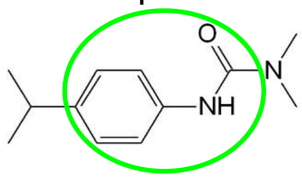
Diuron



Linuron



Isoproturon



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CA assumes that the compounds in the mixture have the same concentration-response relationship (same slope). A more flexible model represents Generalized CA that allows for different slopes, see:

Howard G.J. & Webster T.F. (2009) Generalized concentration addition: A method for examining mixtures containing partial agonists. Journal of Theoretical Biology 259, 469–477.

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^{n} \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

Observed conc. (µg L^{-1})	5.4	160	1.3
p_i	$\frac{5.4}{166.7} = 0.032$	$\frac{160}{166.7} = 0.96$	$\frac{1.3}{166.7} = 0.008$
Algae 72-h EC50, growth (µg L^{-1})	2.7	16	13
13 $\frac{p_i}{EC_{xi}}$	$\frac{0.032}{2.7} = 0.012$	$\frac{0.96}{16} = 0.06$	$\frac{0.008}{13} = 0.0006$

$$(x)^{-1} = \frac{1}{x}$$

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^{n} \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

$$\frac{p_i}{EC_{xi}} \qquad \frac{0.032}{2.7} = 0.012 \qquad \frac{0.96}{16} = 0.06 \qquad \frac{0.008}{13} = 0.0006$$

$$\sum_{i=1}^{n} \frac{p_i}{EC_{xi}}$$

0.012 + 0.06 + 0.0006 = 0.0726

$$ECx_{mixCA} = \frac{1}{0.0726} = 13.77$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^{n} \frac{c_i}{EC_{xi}}$$

c_i: concentration of compound *i*

EC_{xi}: effect concentration *x*% for compound *i*

Observed conc. (µg L^{-1})	5.4	160	1.3
Algae 72-h EC50, growth (µg L^{-1})	2.7	16	13
$\frac{c_i}{EC_{xi}}$	$\frac{5.4}{2.7} = 2$	$\frac{160}{16} = 10$	$\frac{1.3}{13} = 0.1$

$$sumTU = 2 + 10 + 0.1 = 12.1$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^{n} \frac{c_i}{EC_{xi}}$$

c_i: concentration of compound *i*

EC_{xi}: effect concentration *x*% for compound *i*

$$sumTU = 2 + 10 + 0.1 = 12.1$$

Same as $$\frac{\sum_{i=1}^{n} c_i}{ECx_{mixCA}} = \frac{166.7}{13.77} = 12.1$$

Independent action (IA)

$$E(c_{mixIA}) = E(c_1+c_2+...+c_n) = 1-\prod_{i=1}^{n}[1-E(c_i)]$$

$E(c_{mixIA})$: predicted effect of the mixture following IA

$E_{(ci)}$: effect of compound *i* at concentration *c*

<u>Example: Insecticide, herbicide and fungicide</u>

	Diuron	Thiacloprid	Tebuconazole
Observed concentrations (µg L^{-1})	50	10	20

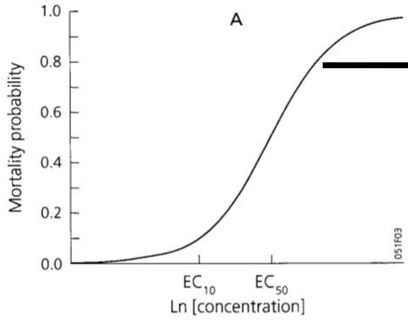
Van Wijngaarden 1996 *Environ. Toxicol. Chem.* 15:1133

Independent action (IA)

$$E(c_{mixIA}) = E(c_1+c_2+...+c_n) = 1-\prod_{i=1}^{n}[1-E(c_i)]$$

<u>Example: Insecticide, herbicide and fungicide</u>

	Diuron	Thiacloprid	Tebuconazole
Observed concentrations (µg L)	50	10	20



Derive $E(c_i)$

$E(c_i)$	0.05	0.2	0.35

Van Wijngaarden 1996 *Environ. Toxicol. Chem.* 15:1133

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + ... + c_n) = 1 - \prod_{i=1}^{n} [1 - E(c_i)]$$

$E(c_i)$	0.05	0.2	0.35
1- $E(c_i)$	0.95	0.8	0.65

$$\prod_{i=1}^{n} [1 - E(c_i)]$$ 0.95 * 0.8 * 0.65 = 0.49

$$E(c_{mixIA}) = 1 - 0.49 = 0.51$$

Computation of IA often hampered by data availability
(mostly only NOEC, LOEC, EC50 reported)

Chevre & Gregoria 2013

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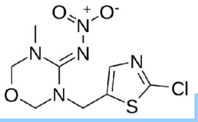
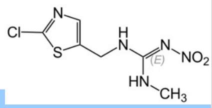
Chèvre N., Gregorio V. 2013: Mixture Effects in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379982

Exercise

$$E(c_{mixIA}) \;=\; E(c_1+c_2+...+c_n) \;=\; 1-\prod_{i=1}^{n}[1-E(c_i)] \qquad sumTU \;=\; \sum_{i=1}^{n}\frac{c_i}{EC_{xi}}$$

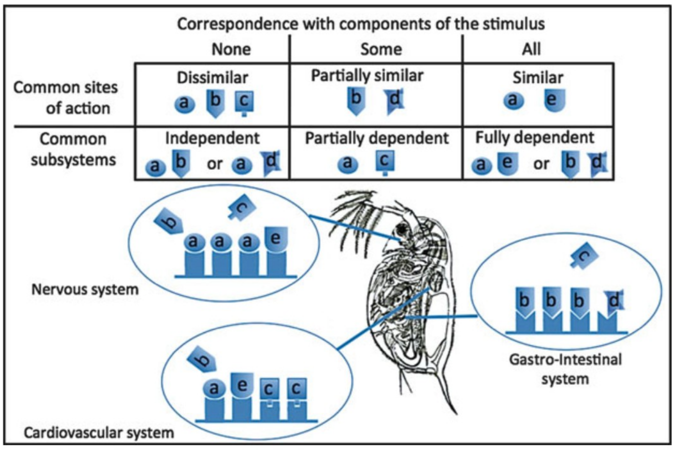
Please calculate the mixture toxicity using CA as sumTU and IA models for the following situation:

	Thiametoxam 	**Clothianidin** 
96-h LC50 Chironomus riparius (µg/L)	100	29
Observed concentrations (µg/L)	5	2.9
E(c_i)	0.02	0.2

Result:
SumTU = 0.15
E(c mixIA) = 21.6%

CA or IA?

- Same target site and MoA → CA
- But MoA can vary between organs



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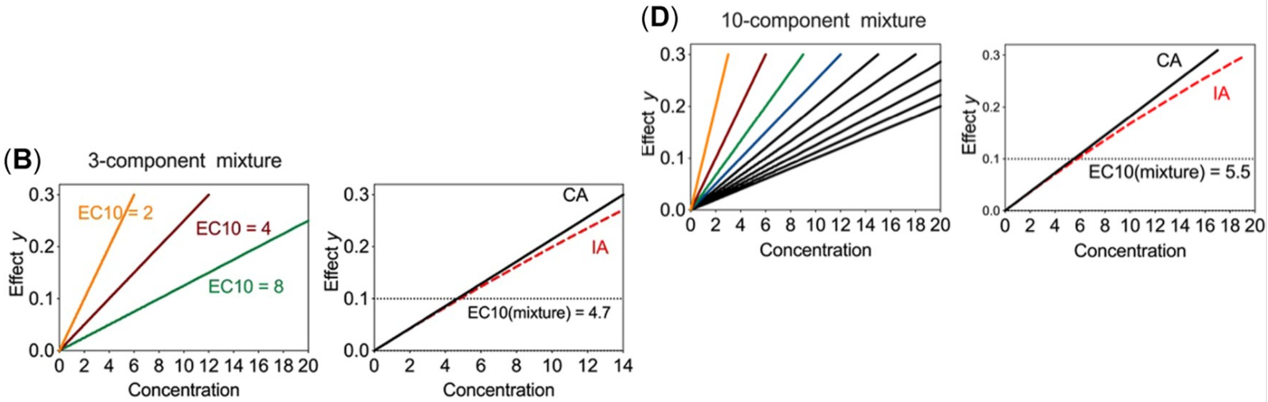
Chevre & Gregoria 2013

Chèvre N., Gregorio V. 2013: Mixture Effects in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379982

CA or IA?

- For equipotent mixtures with low effect levels (i.e. <EC10), often predictions converge
- CA standard because data often too limited for IA



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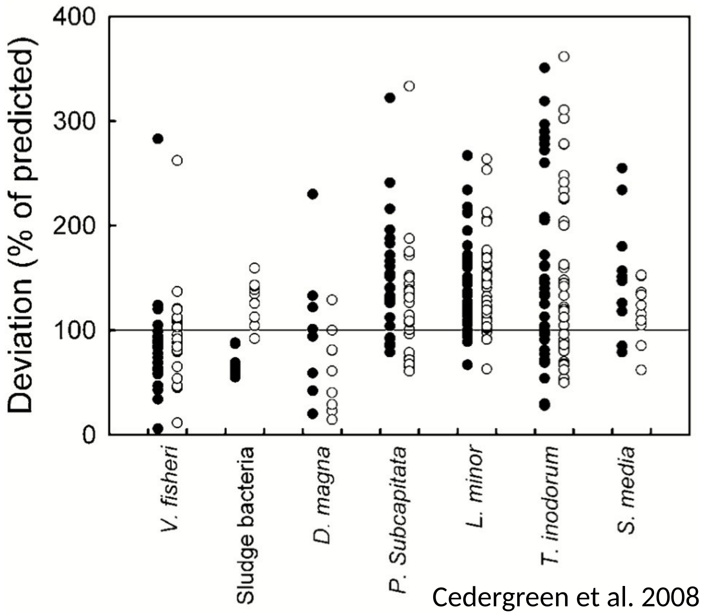
Escher et al. 2020 *Environ. Toxicol. Chem.* 39:2552

Escher, B., Braun, G., & Zarfl, C. (2020). Exploring the Concepts of Concentration Addition and Independent Action Using a Linear Low Effect Mixture Model. Environmental Toxicology and Chemistry, 39(12), 2552–2559. https://doi.org/10.1002/etc.4868

How good is the prediction?

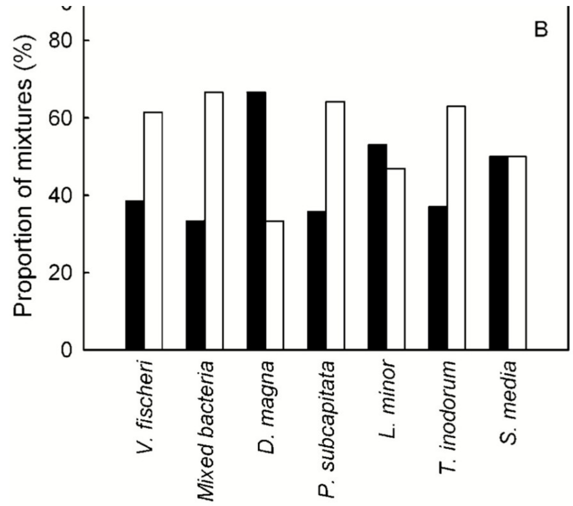
Ratio of CA (black) and IA (white) prediction to observed EC50



Cedergreen et al. 2008 *Environ. Toxicol. Chem.* 27(7): 1621

How good is the prediction?

Proportion of mixtures better described with CA (black) and better described with IA (white)



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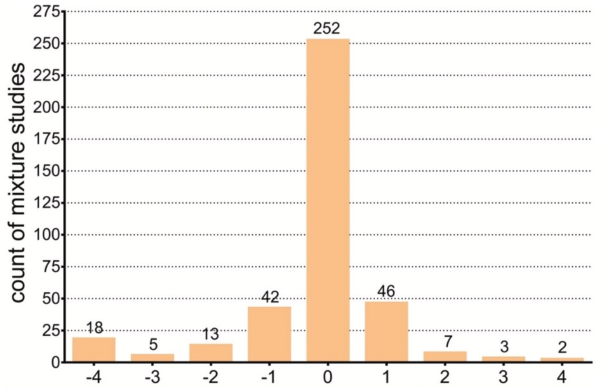
Cedergreen et al. 2008 *Environ. Toxicol. Chem.* 27(7): 1621

The results are based on 158 data sets.

There is slight indication that IA is better than CA (56 of 98 mixtures where better described by IA), but *Daphnia magna* shows the opposite trend.

How good is the prediction?

Observations largely within 1 order of magnitude of mixture prediction



Class	-4	-3	-2	-1	0	1	2	3	4
IPQ range	<- 700%	-700% - -500%	-500% - -300%	- 300% - -100%	-100% - 100%	100% - 300%	300% - 500%	500% - 700%	>700%
	synergisms					antagonism			

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Martin et al. 2021 *Environ. Intern.* 146: 106206

IPQ = Index of Prediction Quality

The formula is given below, where the ratio is the observed to predicted effect. This means that if the observed effect concentration (equivalent to weaker effect) is twice the predicted effect concentration, the IPQ is (+)100%. Conversely, if the observed is half of the predicted effect concentration, the IPQ is -100%.

$$IPQ = \begin{cases} 100*(Ratio - 1) if \quad Ratio \geq 1 \\ 100*\left(1 - \frac{1}{Ratio}\right) if \quad Ratio < 1 \end{cases}$$

How to identify synergism and antagonism?

- Model Deviation Ratio (MDR) often used in Ecotoxicology:

$$\text{MDR} = \frac{\text{Expected/predicted EC with null model}}{\text{Observed EC}}$$

- MDR sets threshold based on effect size (e.g. 0.5 and 2)
- Why not statistical significance?
  - Sample-size dependent: e.g. low effect size (deviation from model) significant for large sample size, high effect size not significant for small sample size

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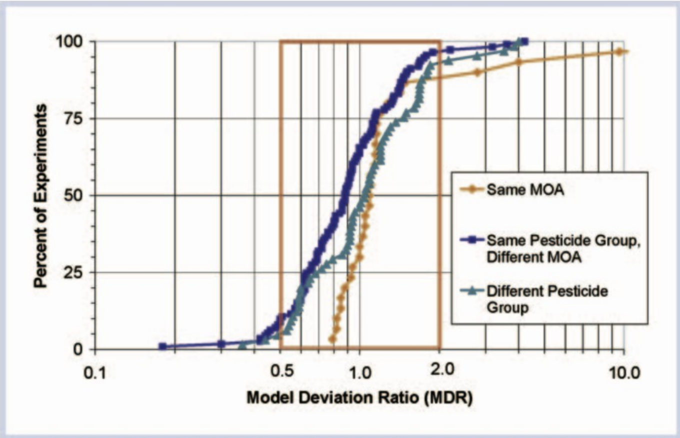
Belden et al. 2007 Integr. *Environ. Assess. Manag. 3* (3): 364

Note that Belden et al. (2007) uses the reciprocal ratio, i.e. predicted to observed. Therefore, MDR > 1 means that the prediction underestimated the effect, i.e. greater toxicity. A value < 1 means that the prediction overestimated the effect.

Note also that if effects were compared directly, what is underestimation and overestimation may again swap.

How good is the prediction?

MDR for CA for pesticide mixtures → Reasonable match for majority of experiments, low fraction of synergism/antagonism



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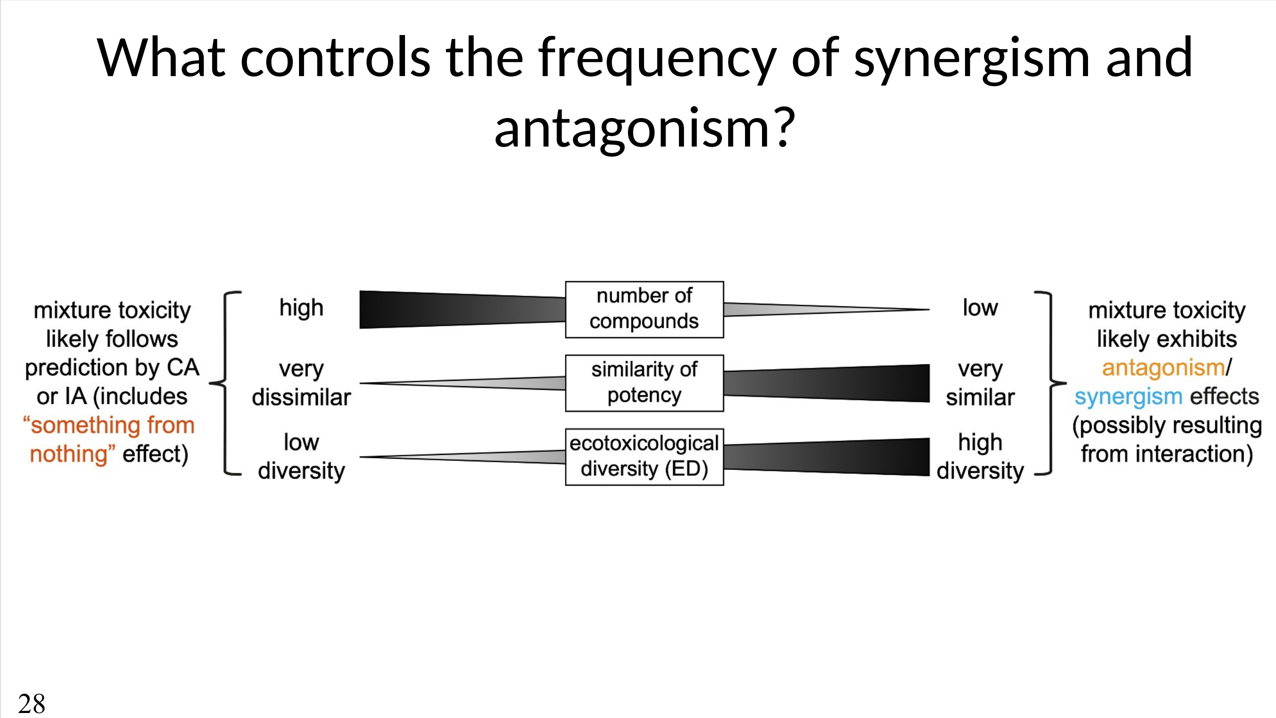
Belden et al. 2007 Integr. *Environ. Assess. Manag. 3* (3): 364

The figure shows the cumulative distribution of MDR for concentration addition experiments grouped by mixtures with the same mode of action ([MOA]; n = 30, indicated by diamond symbol), mixtures of the same pesticide use group and different MOA (n = 113, indicated by square symbol), and mixtures of different pesticide use groups and different MOA (n = 64, indicated by square symbol).

Species tested are shown in Table 1 of Belden et al. (2007)

Table 1. The number of experiments testing concentration addition (CA), independent action (IA), and simple interaction (SI) models for each taxonomic group. $n = 303$

Higher taxonomic group	Predictive model CA	IA	SI
Amphibia	4	—	3
Bivalvia	2	—	—
Chlorophyta	87	10	—
Crustacea	20	4	14
Insecta	16	1	39
Liliopsida	38	19	—
Osteichthyes	38	1	3



Ecotoxicological diversity (ED) refers to the diversity of ecotoxicological mechanisms in terms of distinct Modes of Action exhibited by components of a chemical mixture. Its calculation reflects the distribution of toxic potency across Modes of Action.

The funnel hypothesis states that, where interaction effects occur, their quantitative impact decreases as the number of chemicals in equipotent mixtures increases (Warne & Hawker 1995). This is because potentially interacting chemicals contribute less to the overall mixture toxicity and interaction effects become negligible or cancel out. Consequently, interaction effects found in mixture experiments for binary and ternary mixtures are likely absent in mixtures composed of many chemicals. Experimental studies largely supported the funnel hypothesis, except when the diversity of MoA, i.e., that is the ED, increased with the number of compounds (Rodea-Palomares et al. 2010).

Rodea-Palomares, I., Petre, A. L., Boltes, K., Leganés, F.,

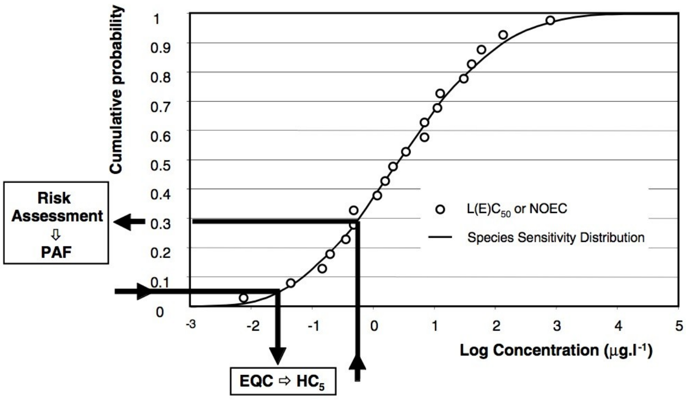
msPAF: Mixture prediction for SSDs

- Mixture prediction (CA and IA/RA) for community level effects
- Instead of *ECx* and *E(c)* values, use *Hcp* and *PAF* values from single-substance SSDs

IA is termed response addition (RA) in this context

SSDs for risk assessment

- Hazardous concentration (HC_p): conc. at which *p*% affected
- Potentially affected fraction PAF: given by concentration



30 Cutoff for community PNEC Posthuma, Traas & Suter 2002

Posthuma, L., Traas, T. P., & Suter, G. W. (2002). General Introduction to species sensitivity distributions. In L. Posthuma, G. W. Suter, & T. P. Traas (Eds.), Species sensitivity distributions in ecotoxicology (pp. 3–10). Lewis.

msPAF: Mixture prediction for SSDs

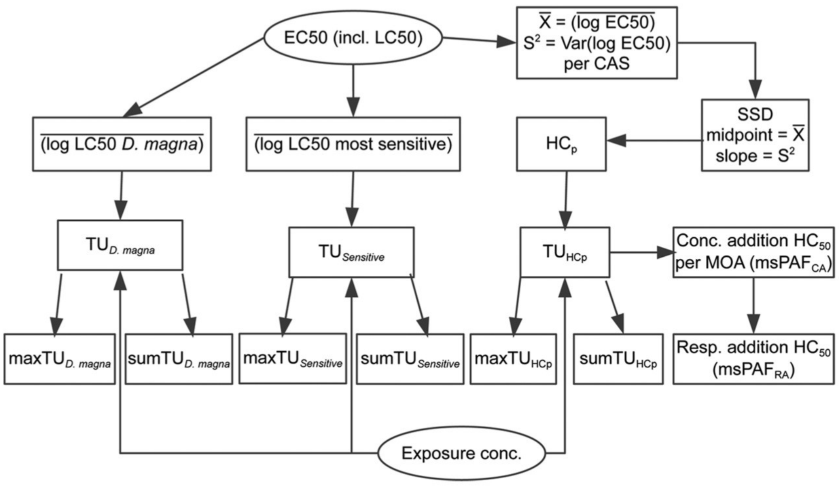
- Mixture prediction (CA and IA/RA) for community level effects
- Instead of ECx and $E(c)$ values, use Hcp and PAF values from single-substance SSDs
- $msPAF_{CA}$: Aggregate Hcp for all compounds with same MoA
- Alternative: $sumTU_{HCp}$

$$msPAF_{RA} = 1 - \prod_{i=1}^{n} \left(1 - msPAF_{CA,i}\right)$$

IA is termed response addition (RA) in this context

Example for community-level mixture indices

IA/RA was only calculated in SSD context due to lack of data



Schäfer et al. 2013 *Environ. Sci. Technol.* 47 (14): 7996

Example is for freshwater invertebrate communities, in case of other organism groups, a different test organism than *Daphnia magna* would be relevant.

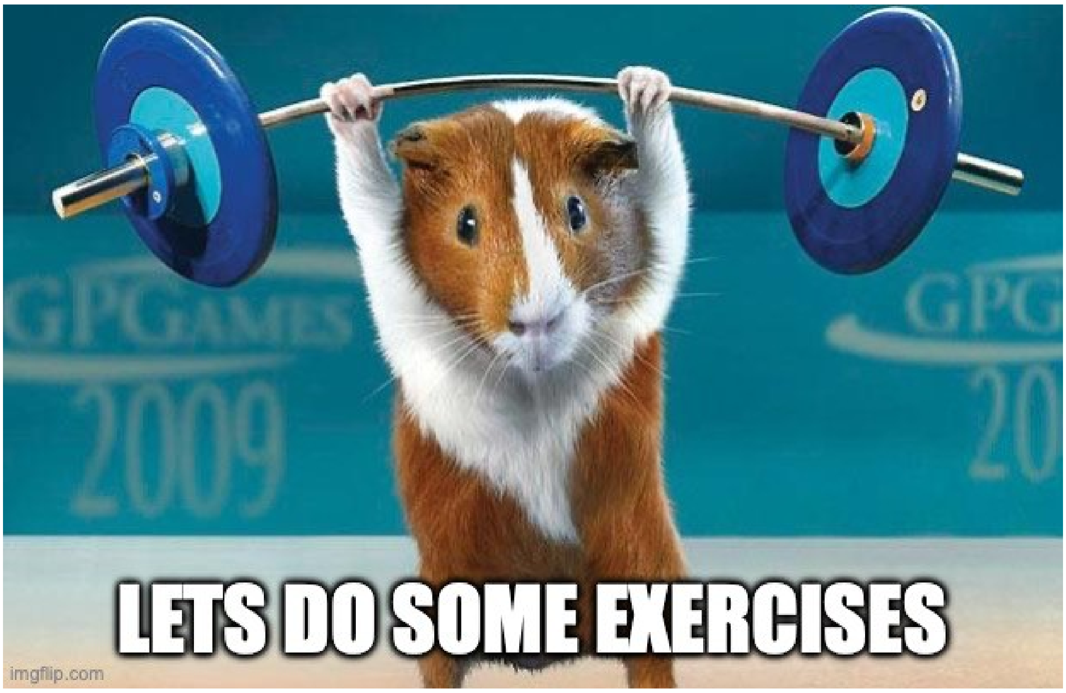
Example for community-level mixture indices

For 4 of 5 data sets *sumTU*$_{HC5}$ best metric, but maxTU and metrics based on *D. magna* or most sensitive test species not much worse

Exposure metric	Rank sum of metric across countries	Number of times metric was among models within a range of 2 to BIC of best-fit model
sumTU$_{HC5}$	14	4
sumTU$_{Sensitive}$	18	3
maxTU$_{HC5}$	21	3
maxTU$_{Sensitive}$	21	3
maxTU$_{D.magna}$	22	2
sumTU$_{D.magna}$	26	3
msPAF$_{RA}$	32	1
maxTU$_{HC50}$	35	1
sumTU$_{HC50}$	36	1

Schäfer et al. 2013 *Environ. Sci. Technol.* 47 (14): 7996

Example is for freshwater invertebrate communities, in case of other organism groups, a different test organism than *Daphnia magna* would be relevant.


LETS DO SOME EXERCISES
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